Geometric Root Calculators

A ninth-order cubic construction and real-degree root solvers

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https://github.com/ivan-fr/pandrosion

Abstract. We study positive-root constructions based on the Pandrosion rectangle. A decentered moving arc gives a globally monotone fifth-order correction. A nested-radical support uses two successive mean proportionals and has exact cubic local order nine with coefficient $7/648$. Its unmodified scalar family converges from every positive start for all real $p > 2$; degree doubling and inversion extend it to every $p \neq 0$. These convergence results and exact cubic order are checked in Lean; general local order at least nine has a separate analytic proof. A complete integer-degree inventory bounds a nested-radical step by $10p + 59$ traces. We compare rectangle supports of orders one, two, three, four, five and nine at a common starting state. A separate power-free subdivision construction encloses real-degree roots with an explicit width and drawing bound. Figures detail the first nested-radical step for $\sqrt[3]{2}$. Historical priority, minimum drawing cost and runtime superiority remain unestablished.

1 Rectangle supports and real-degree roots

A geometric iteration must realize its arithmetic by intersections, parallels and length transfers. We use the same rectangle readout with several supports: MA, AK, centered AD, projective AD, decentered AD and a nested-radical support. Their orders range from one to nine in the cubic comparison; this is a statement about error decay, not an ordering of physical drawing efficiency.

Three results organize the paper:

1. **Decentered support.** One moving support arc gives a fifth-order correction with global monotone convergence for real $p > 2$ (Theorem 3.1). Its mobile support operations match those of centered AD.
2. **Nested radicals.** Two successive mean proportionals give exact cubic order nine (Theorem 4.1) and global convergence for all real $p > 2$ (Theorem 4.2); an explicit adaptation covers every $p \neq 0$ (Proposition 4.3). The general local result is order at least nine. The complete integer-degree inventory is $10p+59$ traces per step (Proposition 5.1).
3. **Direct enclosure.** For a given real degree $p \geq 1$, subdivision and a final rational readout enclose $X^{1/p}$ to prescribed relative tolerance without evaluating a residual power (Proposition 6.1). The conservative bound for $X > 1$ is $15n + 63$ traces after n subdivisions.

The common coordinates and primitive arithmetic are given in Section 2. Sections 3 and 4 develop the radical supports; Section 5 compares the six retained rectangle constructions. The direct enclosure, attribution and proof scopes follow. Classical supports, the MA reference and algebraic equivalence details are retained in the appendices. Throughout, s is a reciprocal-root estimate; the nested-radical section explicitly uses the direct state $u = 1/s$.

[1]Computational assistance: AI was used for derivations, proof development, figures and manuscript preparation. The author is responsible for the claims and their attribution. Section 8 distinguishes exact proofs, numerical checks and drawing inventories.

2 Coordinates and reusable arithmetic

For $X > 0$, let $a = X^{-1/p}$, $t = Xs^p$ and $s^+ = s\Phi(t)$, with $s > 0$. The reciprocal state approaches a; inversion supplies $X^{1/p}$. Write $e = \log(s/a)$ for logarithmic error. The scalar formulas allow real degrees where stated, while the finite power-stage inventories below concern integer $p \geq 3$.

In the rectangle $C = (0,0)$, $B = (W,0)$, $O = (0,H)$, $A = (W,H)$, with $W, H > 0$, define

$$R(q) = (W, H(1-q)), \quad L(q) = (0, H(1-q)), \quad \mathcal{U}_f = \left(\frac{Wf}{f-1}, H\right) \quad (f \neq 1). \tag{1}$$

The line incidences perform multiplication and division:

$$\mathcal{U}_f R(q) \cap OC = L(fq), \qquad \mathcal{U}_f L(b) \cap AB = R(b/f). \tag{2}$$

Substituting the coordinates into a line equation proves both identities. Unit factors and zero values use the stated reuse shortcuts rather than an infinite center.

2.1 Triangle power stage

The rectangle drawings obtain $E = R(s^p)$ directly. Let $B_j = (Ws^j, H(1-s^j)) \in OB$ and $L_j = L(s^j)$. Horizontals transfer between these points and the vertical rails. A parallel to L_1B through B_j meets the left rail at L_{j+1}: its ordinate is $H(1-s^{j+1})$. Repetition gives the integer power endpoint.

2.2 Work convention and finite products

Count a mobile join as J, a constructed parallel as $P_{\parallel}$ and a mobile compass arc as C_{arc}. A parallel macro expands to $2J{+}3C_{\text{arc}}$. Existing intersections, fixed preparation, decisions and the final inversion of reciprocal-state iterations are excluded. Section 6 explicitly counts its input/output inversions when $0 < X < 1$. A trace is one join or one arc after expansion; this unweighted count is not time or energy.

For positive rail values a, b, reuse $R(a)$ when $b = 1$. Otherwise draw $V = CR(b) \cap OA = \mathcal{U}_{1/b}$, then project through $\mathcal{U}_{1/2}, V, \mathcal{U}_{1/2}, \mathcal{U}_{1/4}$. The successive values are $L(a/2), R(ab/2), L(ab/4), R(ab)$: multiplication costs five joins. The centers $\mathcal{U}_{d/[2(1+d)]}$ and $\mathcal{U}_{1/[2(1+d)]}$ multiply a value by a prepared $d > 0$ in two joins. Division and mean-proportional macros are detailed in Section 6.

The name Pandrosion refers to the addressee of Book III of Pappus' *Collection* [1]. Our rectangle follows a modern reconstruction [2]; the accelerated supports are modern constructions, not attributed to the ancient text.

3 A decentered arc with global fifth-order convergence

The moving-support architecture keeps the power endpoint $E = R(s^p)$ and changes the interception support inside the rectangle. Prepare $M = (0, H(1-X^{-1}))$. The slope of ME is $m = H(1-t)/(WX)$. A line through A of slope $k_s = H\kappa/(WX)$ meets the parallel to ME through $P = R(s)$ at T_A. The horizontal through T_A gives P^+, with

$$s^+ = s\frac{k_s}{k_s - m} = s\frac{\kappa}{\kappa - 1 + t}. \tag{3}$$

The decentered arc will supply $\kappa = 1 + g(t)$. The common triangle stage uses horizontals through $B_j = (Ws^j, H(1 - s^j))$ on OB and $L_j = L(s^j)$. Parallels to L_1B carry B_j to L_{j+1}; the pth horizontal gives E. This is the triangle protocol of Section 2. The classical AK, AD and projective supports are documented in Appendix A; MA remains in Appendix B.

3.1 Realizing the positive radical

For $p > 2$, define

$$q(t) = \sqrt{\frac{(p-2)(t^2+1) + (10p+4)t}{12p}}, \quad g(t) = (p-1)q(t), \quad \mathcal{T}_{\text{arc}}(s) = s\frac{1+g(t)}{t+g(t)}. \tag{4}$$

For integer $p \geq 3$, put $\eta = (p-1)\sqrt{(p-2)/(12p)}$, $\zeta = (5p+2)/(p-2)$, $d_0 = W/\eta$ and $\delta = (H/X)\sqrt{\zeta^2 - 1}$. Prepare

$$G_0 = (W, H + H\zeta/X), \quad O_{\text{arc}} = (W - d_0 + \delta, H - H/(X\eta)), \quad x = W - d_0.$$

Draw the arc centered at O_{arc} with radius $EG_0 = (H/X)(t+\zeta)$ and take the lower intersection D_{arc} with the prepared vertical. Its vertical distance below the center is

$$\sqrt{EG_0^2 - \delta^2} = \frac{H}{X}\sqrt{t^2 + 2\zeta t + 1}.$$

The support AD_{arc} consequently gives $\kappa = 1+g(t)$ in (3). All preparation parameters use rational operations and square roots of positive quantities. The radicand stays positive for $t > 0$.

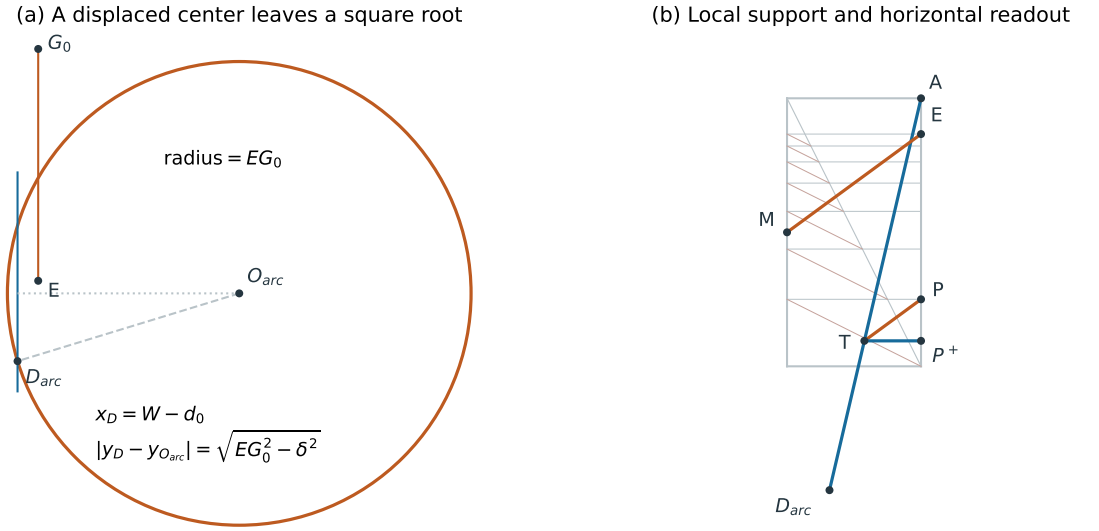


Figure 1: The decentered-arc device for $p = 7, X = W = 2, H = 4, s = 3/4$. The full circle geometry is shown separately from the local report, at independently chosen equal-axis scales. The displacement δ makes the vertical circle intersection perform a square root. The local panel retains the complete support and distinguishes the reported state.

3.2 Global monotone convergence of the decentered map

The correction is defined for every positive residual and uses the positive radical, with a mobile radius at each step.

Centered AD and decentered AD use the same mobile support operations: one arc and one join. Their fixed preparations and coordinate scales differ. This is the precise sense in which order five is obtained at the mobile support cost of order three.

Theorem 3.1. *For real $p > 2$, $X > 0$ and every $s_0 > 0$, (4) converges to a monotonically without crossing it. Its logarithmic error satisfies*

$$e^+ = \frac{(p-2)(p^2-1)(3p+1)}{1440} e^5 + O(e^7). \tag{5}$$

Proof. The radicand is positive and $q(1/t) = q(t)/t$, so the correction is reciprocal. Write

$$\mathcal{E}(t) = \frac{\log t}{p} + \log(1 + g(t)) - \log(t + g(t)).$$

Define the auxiliary polynomials

$$\begin{aligned} A_*(t) &= (p-1)(t^2+1) + (10p+2)t, \\ B_*(t) &= (t+1)(6pt - (t-1)^2), \\ K_*(t) &= (p^2-5p-2)(t-1)^2 + 12p(3p+1)t. \end{aligned}$$

Differentiation and factorization give

$$\frac{d\mathcal{E}}{d\log t} = \frac{(p-2)(p-1)^2(qA_* - B_*)}{12p^2 g(1+g)(t+g)}, \tag{6}$$

$$q^2 A_*^2 - B_*^2 = \frac{p+1}{12p}(t-1)^4 K_*. \tag{7}$$

Here $A_* > 0$. If $B_* \le 0$, then $qA_* - B_* > 0$. Otherwise $(t-1)^2 < 6pt$. When $p^2 - 5p - 2 \ge 0$, $K_* > 0$ directly; otherwise

$$K_* = 6p^2(p+1)t + (p^2-5p-2)((t-1)^2 - 6pt) > 0.$$

Equation (7) now gives $qA_* > B_*$ for $t \ne 1$. Therefore $\mathcal{E}$ is strictly increasing with $\mathcal{E}(1) = 0$: the new logarithmic error has the old sign. Meanwhile $\Phi(t) - 1 = (1-t)/(t+g)$ moves the state toward the root. The orbit is monotone and bounded between s_0 and a. Its positive limit is a fixed point by continuity, and $\Phi(t) = 1$ implies $t = 1$.

Near one, rationalize $qA_* - B_*$ using (7). Since $q = 1$, $g = p-1$, $A_* = B_* = 12p$ and $K_* = 12p(3p+1)$ at $t = 1$, this gives

$$\lim_{t\to 1} \frac{d\mathcal{E}/d\log t}{(t-1)^4} = \frac{(p-2)(p^2-1)(3p+1)}{288p^5}.$$

Substituting $t = e^{pe}$ and integrating yields (5); reciprocity makes the analytic logarithmic map odd. □

At $p = 3$, the leading logarithmic coefficient is $1/18$. At $p = 2$ the scalar formula gives an exact square-root correction, but the displayed preparation is singular and must be replaced by a mean-proportional construction.

4 A nested-radical support with cubic order nine

The next support uses two successive mean-proportional constructions, with the output of the first entering the diameter of the second. It is a new member of the collection studied in this paper, without a claim of historical priority. Unlike the direct enclosure of Section 6, this iteration still requires a power residual. We use a direct-root state $u = 1/s$ and target $r = X^{1/p}$ in this section. The residual remains $t = X/u^p = Xs^p$, so the change of state does not change the earlier rail encoding.

4.1 Formula, domain and rectangle support

For $p > 2$, write $h = 1/p$ and define

$$a_9 = \frac{1-2h}{12}, \qquad b_9 = \frac{(h+1)(h+3)}{30}, \qquad c_9 = \frac{121-185h-67h^2-h^3}{840(h+3)}, \tag{8}$$

$$w = \frac{(t-1)^2}{t}, \qquad k_9 = \sqrt{1+\frac{b_9 w}{1+c_9 w}}, \qquad q_9 = \sqrt{t\left(1+\frac{a_9 w}{k_9}\right)}. \tag{9}$$

All square roots mean their positive branches. Set

$$C_p(t) = \frac{t+(p-1)q_9(t)}{1+(p-1)q_9(t)}, \qquad F_{p,X}(u) = uC_p(X/u^p). \tag{10}$$

For $0 < h < 1/2$, all three coefficients are positive. In particular the numerator of c_9 is decreasing on this interval and has value $93/8 > 0$ at $h = 1/2$. Thus every division and radical in (9) is defined for every $t > 0$. Also

$$w(1/t) = w(t), \quad q_9(1/t) = q_9(t)/t, \quad C_p(1/t) = 1/C_p(t). \tag{11}$$

To use the existing rectangle, take $g_9 = (p-1)q_9$ and $\kappa = 1 + g_9$ in (3). One possible support point is

$$D_9 = \left(0, H - \frac{H}{X}(1+g_9)\right).$$

The parallel to ME through $P = R(s)$ meets AD_9, and its horizontal readout gives

$$s^+ = s\frac{1+g_9}{t+g_9}, \qquad 1/s^+ = F_{p,X}(u).$$

The arithmetic after obtaining t consists of rational operations and two square roots. Each root is a mean proportional; the general formula uses $\sqrt{1\cdot(1+b_9w/(1+c_9w))}$ and $\sqrt{t(1+a_9w/k_9)}$. For integer p, the residual is constructible by the earlier product protocols. For arbitrary real p, evaluating u^p is an analytic primitive here, not a finite ruler-and-compass construction.

4.2 First complete step for the cube root of two

At $p = 3$, the support simplifies to

$$q_9(t) = \sqrt{t + \frac{(t-1)^2}{108}\sqrt{\frac{t^2+52t+1}{t^2+4t+1}}}, \qquad u^+ = u\frac{t+2q_9(t)}{1+2q_9(t)}. \tag{12}$$

Here is a construction of the first step for $X = 2$, $u_0 = 1$; the small auxiliary diagrams have their own local point names and may be translated or uniformly enlarged. The unknown cube root is never used to place a point.

1. **Rational preparation.** The residual is $t = 2$. Prepare $109/13 = 8 + 5/13$ and $1/108$ with subdivisions and similar triangles. These are rational operations, not extra root extractions.
2. **First semicircle.** Place adjacent segments of lengths 1 and $109/13$. Draw their diameter semicircle and the perpendicular at the junction. Its height is $\ell = \sqrt{109/13} = 2.895620034572\ldots$. The cubic auxiliary ℓ is not the general k_9 of (9).
3. **Prepare the second diameter.** A similar-triangle division gives $d = \ell/108 = 0.026811296616\ldots$. Add it to two units.
4. **Second semicircle.** Use adjacent lengths 1 and $2 + d$. The perpendicular height is $q_9 = \sqrt{2+d} = 1.423661229583\ldots$.
5. **Move the support.** Choose $W = 6, H = 4$, so $C = (0,0)$, $B = (6,0)$, $O = (0,4)$, $A = (6,4)$ and $M = (0,2)$. Here $s_0 = 1$, hence $P_0 = E = B$. Place D_9 below M by $4q_9$: $D_9 = (0, 2 - 4q_9)$.
6. **Intersect and report.** Draw AD_9 and MB; their intersection is T_9. The horizontal through T_9 meets AB at P_1. The rectangle reports $s_1 = AP_1/AB$.
7. **Invert the reading.** A final similar-triangle construction produces $u_1 = AB/AP_1$. This is the direct-root estimate.

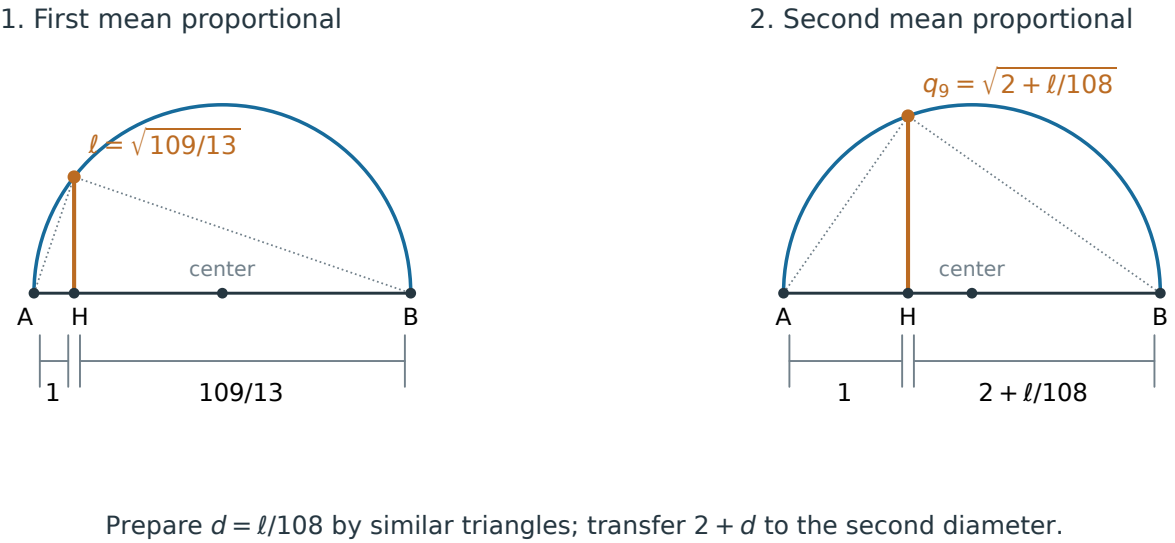


Figure 2: The two successive mean proportionals for $X = 2, u_0 = 1$. Each panel uses equal axis scales and its own magnification. The first height supplies the rationally scaled addition d used by the second diameter. The orange perpendiculars are the constructed lengths; decimal values are checks on ideal incidences.

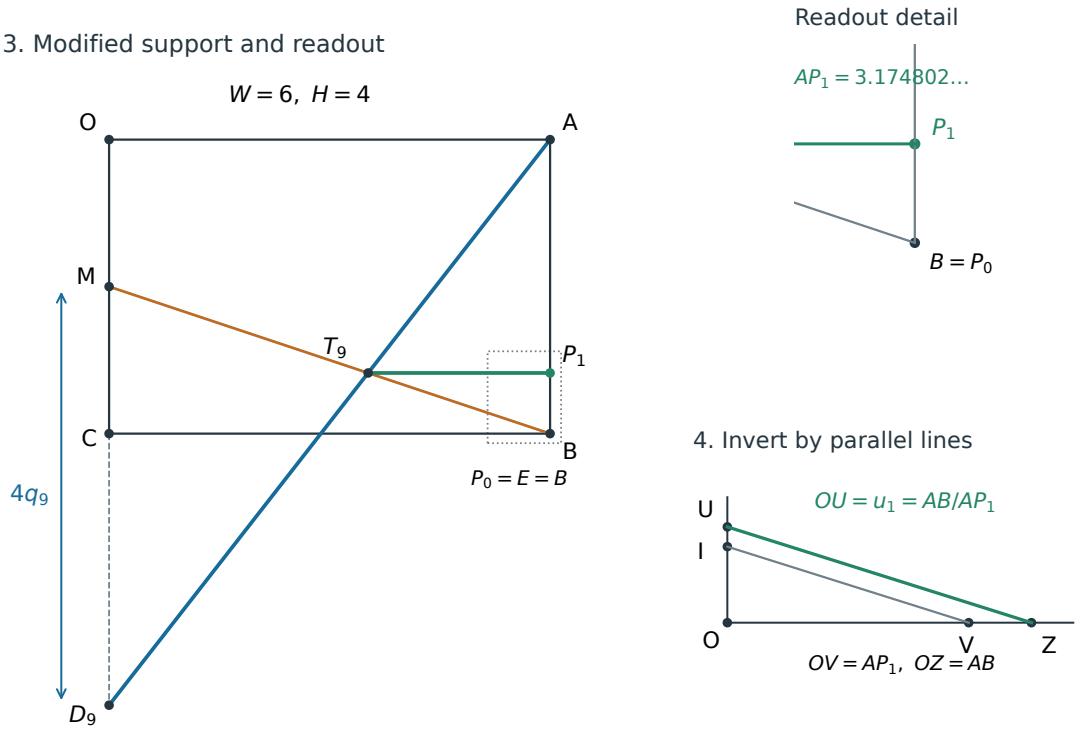


Figure 3: The modified Pandrosion support for the same first step. The complete rectangle and support AD_9 are drawn at equal scales. The upper inset enlarges the horizontal readout, with its own equal scale; the lower panel constructs the reciprocal reading by parallel lines. Because $s_0 = 1$, $P_0 = E = B$ and the transport line is MB itself. The extension below C is part of the construction.

The incidences give

$$\begin{aligned} T_9 &= (3.524406494288\ldots,\ 0.825197835237\ldots), \\ s_1 &= 0.79370054119068191329\ldots, \\ u_1 &= 1.25992102575592908435636220628 1\ldots. \end{aligned}$$

The relative error is approximately $-1.9159092613\times10^{-8}$. These digits describe exact geometric incidences evaluated numerically; they are not readable to that precision from a manual drawing. Later steps must recompute $t = 2/u^3$: the constants 109 and 13 are specific to this first step. The seven grouped instructions, and the twelve screens of the companion interactive display, are pedagogical steps rather than primitive trace counts.

4.3 Local contact and exact cubic order

The support that would produce an exact correction is

$$q_* = \frac{t - t^h}{(p-1)(t^h - 1)}, \qquad q_*(1) = 1.$$

This is used only to derive and prove the method. Put $z = \log t$ and interpret the quotients continuously at zero. Then

$$Y := \frac{q_*^2}{t} = \left[\frac{\sinh((1-h)z/2)/((1-h)z/2)}{\sinh(hz/2)/(hz/2)}\right]^2, \qquad w = 2(\cosh z - 1).$$

Expansion gives

$$\begin{aligned} Y &= 1 + a_9 w + A_2 w^2 + A_3 w^3 + O(w^4), \\ A_2 &= -\frac{(1-2h)(h+1)(h+3)}{720}, \\ A_3 &= \frac{(1-2h)(h+1)(2h^3+8h^2+13h+31)}{60480}. \end{aligned}$$

Meanwhile the proposed support satisfies

$$\frac{q_9^2}{t} = 1 + a_9 w - \frac{a_9 b_9}{2} w^2 + a_9 \left(\frac{b_9 c_9}{2} + \frac{3b_9^2}{8}\right) w^3 + O(w^4).$$

The coefficients in (8) match both A_2 and A_3 exactly. Therefore $q_9 - q_* = O(z^8)$. The derivative of $(t+g)/(1+g)$ with respect to g is $(1-t)/(1+g)^2 = O(z)$, so $C_p(t) - t^{1/p} = O(z^9)$. For each fixed real $p > 2$, the iteration has local order at least nine. This coefficient-matching argument is an analytic proof with exact polynomial checks, not a Lean formalization of the general local order. We do not assert a nonzero ninth-order coefficient for every real $p > 2$.

For the cubic specialization a short positive certificate proves more. Let $v = t^{1/3} > 0$, solely for this proof, and set

$$\begin{aligned} D_v &= v^6 + 4v^3 + 1, \\ H_v &= v^8 + 10v^7 + 55v^6 + 268v^5 + 466v^4 + 268v^3 + 55v^2 + 10v + 1, \\ Z_v &= (v^2+v+1)^2 \sqrt{(v^6 + 52v^3 + 1)/D_v}. \end{aligned}$$

The exact identity

$$(v^2+v+1)^4(v^6+52v^3+1) - 729v^4 D_v = (v-1)^6 H_v \tag{13}$$

implies $Z_v \geq 27v^2$ and gives

$$q_9^2 - \left(\frac{v^2+v}{2}\right)^2 = \frac{(v-1)^8 H_v}{108 D_v (Z_v + 27v^2)}. \tag{14}$$

Every denominator is positive. Since $H_1 = 1134$,

$$q_9 - \frac{v^2+v}{2} = \frac{7}{432}(v-1)^8 + O((v-1)^9).$$

Theorem 4.1 (Exact cubic order nine)**.** *For $X > 0$, $r = X^{1/3}$ and $\epsilon = u/r - 1$, the cubic update satisfies*

$$\lim_{u \to r,\ u \neq r} \frac{F_{3,X}(u)/r - 1}{(u/r-1)^9} = \frac{7}{648}.$$

Its local order is exactly nine. This limit is formalized in Lean.

Proof. In the normalized state $y = u/r$, one has $v = 1/y$. The exact difference $C_3(v^3) - v$ equals $2(1 - v)(q_9 - (v^2 + v)/2)/(1 + 2q_9)$. Substitution in (14), followed by $v = 1/y$, factors $F_{3,1}(y) - 1$ as $(y - 1)^9$ times a continuous positive factor with value $7/648$ at one. Scaling gives $F_{3,r^3}(ry) = rF_{3,1}(y)$ and proves the stated limit. □

Analyticity and reciprocity additionally give the written logarithmic expansion $e^+ = (7/648)e^9 + O(e^{11})$, where $e = \log(u/r)$. The Lean theorem above certifies the relative-error limit; the stronger displayed remainder is an analytic consequence, not a separate audited asymptotic theorem.

4.4 Global convergence for every real degree greater than two

The local cancellation does not prove global convergence. The following independent sign argument covers the entire positive axis.

Theorem 4.2 (Global monotone nested-radical iteration)**.** *For every real $p > 2$, $X > 0$ and $u_0 > 0$, the iteration $u_{n+1} = F_{p,X}(u_n)$ is defined, stays positive and converges to $r = X^{1/p}$. If $u_0 \neq r$, it moves strictly toward r without crossing it. This theorem is formalized in Lean for the original formula (8)–(10).*

Proof. We give the reduction to positive polynomial certificates; their complete integer coefficients are included in the accompanying Lean source. Define

$$\begin{aligned}
\mathsf{E} &= 840p^2(3p + 1),\\
\mathsf{C} &= 121p^3 - 185p^2 - 67p - 1,\\
\mathsf{D} &= 373p^3 + 235p^2 + 129p + 27,\\
\mathsf{U} &= \mathsf{E} + \mathsf{C}w, \qquad \mathsf{V} = \mathsf{E} + \mathsf{D}w, \qquad L = \sqrt{\mathsf{U}/\mathsf{V}},\\
\mathsf{R} &= 1 + \frac{(p - 2)wL}{12p}.
\end{aligned}$$

Here $L = 1/k_9$ and $q_9 = \sqrt{t\mathsf{R}}$; the equivalence is itself formally checked. For $p > 2, w \geq 0$, the quantities $\mathsf{U}, \mathsf{V}, L, \mathsf{R}$ are positive. Put

$$\begin{aligned}
\mathsf{J} &= 4\mathsf{UV} - p\mathsf{E}(\mathsf{D} - \mathsf{C})w,\\
\mathsf{A} &= 1 + \frac{(p - 1)wL}{12p}, \qquad \mathsf{B} = 1 - \frac{\mathsf{J}wL}{24p\mathsf{UV}}.
\end{aligned}$$

The two-region certificate establishes

$$0 < w \leq 30p \implies \mathsf{RA}^2 > (1 + w/4)\mathsf{B}^2, \tag{15}$$

$$w \geq 30p \implies \mathsf{B} < 0. \tag{16}$$

For reproducibility, the main elimination uses

$$\begin{aligned}
\mathsf{K} &= -1728p^3\mathsf{UV}^3 + 48p(p - 1)(3p - 5)\mathsf{U}^2\mathsf{V}^2w - 3p\mathsf{J}^2(4 + w)w,\\
\mathsf{M} &= 576p^2(3p - 4)\mathsf{UV} + 144p^2\mathsf{J}(4 + w) + 4(p - 2)(p - 1)^2\mathsf{U}^2w^2.
\end{aligned}$$

There is a polynomial $\mathsf{P}(p, w)$ of degree nine in w such that

$$(\mathsf{V}^2\mathsf{M})^2\mathsf{U} - \mathsf{K}^2\mathsf{V} = (p + 1)w^3\mathsf{VP}(p, w).$$

Its exact Bernstein certificate is

$$(30p)^9\mathsf{P}(p, w) = \sum_{j=0}^{9} w^j(30p - w)^{9-j}T_j(p - 2), \tag{17}$$

where every nonzero coefficient of each T_j is positive, and each $T_j(p-2) > 0$ for $p > 2$. The certificate contains 345 positive terms. Positive-coefficient expansions of M in $(p-2, w)$, and of J and $w^2\mathsf{J}^2 - 576p^2\mathsf{UV}^3$ in $(p-2, w-30p)$, provide the remaining signs. Their respective term counts are 49, 24 and 112. Rationalization using $L^2\mathsf{V} = \mathsf{U}$ proves (15) and (16). These are exact identities and inequalities checked by Lean, rather than sampled signs.

Since $(t+1)^2 = 4t(1+w/4)$ and $q_9^2 = t\mathsf{R}$, the two regions give

$$2q_9\mathsf{A} - (t+1)\mathsf{B} > 0 \quad (t > 0,\ t \neq 1). \tag{18}$$

Define $\Delta_p(t) = \log C_p(t) - \log(t)/p$. Exact differentiation yields

$$\Delta_p'(t) = -\frac{(p-1)(p-2)\,[2q_9\mathsf{A} - (t+1)\mathsf{B}]}{2pq_9[t+(p-1)q_9][1+(p-1)q_9]} < 0 \quad (t \neq 1). \tag{19}$$

Together with $\Delta_p(1) = 0$ and $C_p(t) - 1 = (t-1)/(1+(p-1)q_9)$, this proves

$$t > 1: \quad 1 < C_p(t) < t^{1/p}, \qquad 0 < t < 1: \quad t^{1/p} < C_p(t) < 1.$$

Consequently $u < F_{p,X}(u) < r$ below the root and $r < F_{p,X}(u) < u$ above it. The orbit is monotone and bounded. Its limit is positive; continuity makes it a fixed point, and $C_p(t) = 1$ forces $t = 1$. Thus its limit is r. □

4.5 Extension to every nonzero real degree

The unmodified radicals must not be asserted to work at all smaller degrees. For $p = 3/2, t = 100$, one has $c_9 = -25/2376$ and

$$1 + c_9 w = -1/32, \qquad 1 + \frac{b_9 w}{1 + c_9 w} = -15947/25 < 0.$$

Lean also checks this exact counterexample and the absence of a real square root of that negative radicand.

Proposition 4.3 (Degree doubling and inversion)**.** *For every real $p \neq 0$, every $X > 0$ and every prescribed visible start $u_0 > 0$, there is an explicit extension of* (10) *converging globally to $X^{1/p}$. The extension is formalized in Lean.*

Proof. Choose an integer $k \geq 0$ with $m = 2^k$ and $P = m|p| > 2$. For $p > 0$, initialize $v_0 = u_0^{1/m}$, iterate $F_{P,X}$ and report $u_n = v_n^m$. For $p < 0$, use $v_0 = (1/u_0)^{1/m}$ and report $u_n = 1/v_n^m$. In both cases the visible initial value is exactly u_0. Theorem 4.2 gives $v_n \to X^{1/P}$, so the visible limit is $X^{m/P}$ or $X^{-m/P}$, respectively, equal to $X^{1/p}$. The initialization uses k square roots; the report uses integer powers and, if needed, an inversion. For $p > 2$ take $k = 0$ to retain the raw formula. □

The extension also preserves monotone approach without crossing: positive integer powers preserve inequalities and inversion reverses them together with the direction of approach. It still requires residual powers of real degree P. The statement concerns $X > 0$; it does not address signed roots of negative inputs. Degree zero is excluded because it defines no unique root.

4.6 What is different, and what remains unestablished

At $p = 3, t = 2$,

$$q_9^2 = 2 + \frac{\sqrt{109/13}}{108}, \qquad N_{\mathbb{Q}(\sqrt{109/13})/\mathbb{Q}}(q_9^2) = \frac{606419}{151632}.$$

The norm is not a rational square, while $109/13$ is not a rational square. Thus q_9 has degree four over $\mathbb{Q}$. The correction $(2+2q_9)/(1+2q_9)$ is an invertible homography of q_9 and also has degree four. In particular this cubic correction is not a rational function of t with rational coefficients. It cannot be expressed in this normalization using only one square root over $\mathbb{Q}$. This algebraic-degree argument is written and checked with exact rational arithmetic; it is not among the Lean theorems.

The argument does not exclude generalized algebraic or Hermite–Padé approximants, transformed targets or all changes of variable. Algebraic approximants of arbitrary degree are established generalizations of Padé approximation [3]; the coefficient matching above belongs to that broad methodology. We have not identified the exact construction with a prior formula, but a bounded comparison cannot establish historical priority. As in Appendix C, different explicit formulas need not define different local analytic conjugacy classes.

Table 1: Scope of the new nested-radical results. "Formal" means checked in the project Docker/Lean environment.

Claim	Domain	Evidence
Global monotone convergence	Raw formula: $p > 2$, $X, u_0 > 0$	Formal theorem and sign certificates.
All real nonzero degrees	Degree doubling / inversion, $X, u_0 > 0$	Formal extension; raw formula is modified.
Exact order nine	Cubic, every positive target	Formal relative-error limit $7/648$.
Order at least nine	Fixed real $p > 2$	Written expansion and exact coefficient checks.
First geometric step	$p = 3, X = 2, u_0 = 1$	Explicit construction; independent 110-digit incidence check.
Complete mobile inventory	Integer $p \geq 3$, supplied $R(s)$	Written bound $10p+59$; independent incidence checks (Section 5.1).
Historical novelty, minimum cost or speed advantage	No asserted domain	Not established.

Counting the two radicals alone would omit rational arithmetic, residual powers, support placement and report. Section 5.1 now inventories all those operations for the uniform integer-degree protocol, giving $10p + 59$ traces, including 89 at $p = 3$. The cubic order-nine result concerns local error decay; the complete counts in Section 5 provide a separate comparison of the specified drawing protocols. No minimum-cost, runtime or hardware advantage is asserted.

5 Orders, iteration counts and geometric work

The retained constructions share $P = R(s)$ and a horizontal readout. For the first five rows, the power stage uses $2p - 1$ parallels and the join L_1B; transport adds ME and two parallels. Moving AD adds one arc and one join; projective AD adds two joins. All counts use Section 2.2's convention and are upper bounds for specified protocols.

5.1 Complete uniform inventory of the nested support

Use the reciprocal state s throughout, so that $t = Xs^p$ and the final root inversion is performed only when reporting the output. Fixed preparation includes universal constants and lengths depending on p, X, W, H, never on the current state. In particular a_9, b_9, c_9 and $\lambda = (p - 1)/X$ are fixed; every residual-dependent length below is reconstructed at each step.

Obtain $E = R(s^p)$ by the triangle stage, then $R(t)$ by the two-join fixed scale X. Stop if $t = 1$. For a nonterminal step use the exact identity

$$w = t + \frac{1}{t} - 2 = \frac{(t-1)^2}{t} > 0. \tag{20}$$

This requires one five-join division and two rail sums/differences, instead of separately squaring $t - 1$. Construct successively

$$K = 1 + \frac{b_9 w}{1 + c_9 w}, \qquad k_9 = \sqrt{1 \cdot K}, \qquad R_9 = 1 + \frac{a_9 w}{k_9}, \qquad q_9 = \sqrt{t R_9}.$$

The two means use the semicircle macro of Section 6: each costs at most $2J + P_{\|} + 3C_{\mathrm{arc}}$, including diameter preparation and transfer to the rail.

For the support, scale q_9 to λq_9 in two joins. Its rail length is $H\lambda q_9$. One compass arc centered at the prepared point $M = (0, H(1 - X^{-1}))$ transfers this length downward on the left rail, giving

$$D_9 = (0, H(1 - X^{-1}) - H\lambda q_9) = (0, H - H(1 + (p - 1)q_9)/X).$$

Join AD_9 and ME, draw the parallel to ME through $P = R(s)$, and return horizontally from its intersection with AD_9. No separate scalar construction of g_9 or $1 + g_9$ is needed.

Table 2: Complete per-step upper inventory from $R(s)$ to $R(s^+)$, integer $p \geq 3$. Each sum or nonnegative difference is one rail arc; general division is five joins, and each fixed scale is two joins.

Stage	J	$P_{\|}$	C_{arc}
Triangle power endpoint $E = R(s^p)$	1	$2p - 1$	0
Residual $t = Xs^p$	2	0	0
Width $w = t + 1/t - 2$	5	0	2
Inner argument K (two scales, one division, two sums)	9	0	2
First mean $k_9 = \sqrt{1 \cdot K}$	2	1	3
Outer argument R_9 (scale, division, sum)	7	0	1
Second mean $q_9 = \sqrt{tR_9}$	2	1	3
Scale λq_9 and transfer from M to D_9	2	0	1
Support join AD_9	1	0	0
Join ME, transport parallel and horizontal readout	1	2	0
Total	32	$2p + 3$	12

Proposition 5.1 (Complete nested-radical work bound)**.** *For integer $p \geq 3$, $X > 0$ and a supplied state $R(s)$ with $s > 0$, the protocol above realizes the original nested-radical correction with at most*

$$32J + (2p+3)P_{\|} + 12C_{\mathrm{arc}} = (4p+38)J + (6p+21)C_{\mathrm{arc}}, \tag{21}$$

namely $10p + 59$ unweighted traces per nonterminal step. The cubic bound is $50J + 39C_{\mathrm{arc}} = 89$ traces.

Proof. The rail identities establish every division and fixed scale. Both mean-proportional inputs are positive; the prescribed circle intersection and transferred length give the positive square root. The displayed point D_9 gives $\kappa = 1+(p-1)q_9$ in (3), hence precisely the stated correction. Adding the ten rows gives (21); substituting $P_{\|} = 2J + 3C_{\mathrm{arc}}$ proves the expanded count. Unit shortcuts, reused objects and an exact-root stop can only reduce these upper allowances. □

The count uses the uniform formula (9), including at $p = 3$; it does not treat the first illustration's $109/13$ as a constant for later iterates. Fixed preparation, decisions and the final reciprocal report are excluded in the same way as for the other rows. If charged, final inversion adds at most $5J$ once; supplying u_0 instead of s_0 can add another $5J$ initialization. The all-real convergence theorem does not supply a finite real-power construction or a cost for the adapted degree's initialization/readout. This is a written incidence inventory, not a minimum-cost theorem or Lean cost semantics.

5.2 Comparison at the same state and accuracy

Table 3: Mobile work with the triangle power stage, for integer $p \geq 3$. The scalar scope column concerns positive states and inputs. The nested-radical row uses the uniform protocol of Proposition 5.1.

Construction	Order	J	$P_{\parallel}$	C_{arc}	Scope
MA rectangle	1	2	$2p+1$	0	$X > 1$
AK rectangle	2	2	$2p+1$	0	Global
AD rectangle	3	3	$2p+1$	1	Global
Projective AD	4	4	$2p+1$	0	Global
Decentered AD	5	3	$2p+1$	1	Global
Nested-radical support	$9^{\dagger}$	32	$2p+3$	12	Global

Here $9^{\dagger}$ is exact for $p = 3$; the general local result is order at least nine for each fixed real $p > 2$. The unmodified support is globally monotone for $p > 2$, and the adapted extension covers $p \neq 0$ (Section 4).

For the cubic case, a parallel macro counts as five traces. The six per-step bounds are therefore $37, 37, 39, 39, 39, 89$. Table 4 uses the same $X = 2, s_0 = 3/4$ for every method; the new direct state starts at $u_0 = 4/3$, rather than the different $u_0 = 1$ used in the construction illustration. The stopping test is on the same reciprocal relative error for all rows.

Table 4: Steps / summed trace upper bounds to reach $|s/a - 1| \leq 10^{-d}$, with $p = 3, X = 2, s_0 = 3/4$. Counts agree at 250 and 500 decimal digits. Each total multiplies the per-step protocol bound by the number of steps. All rows exclude fixed preparation and final inversion.

Construction	Order	$d = 6$	$d = 12$	$d = 30$	$d = 60$
MA rectangle	1	8 / 296	17 / 629	44 / 1628	90 / 3330
AK rectangle	2	3 / 111	4 / 148	5 / 185	6 / 222
AD rectangle	3	2 / 78	3 / 117	3 / 117	4 / 156
Projective AD	4	2 / 78	2 / 78	3 / 117	3 / 117
Decentered AD	5	1 / 39	2 / 78	2 / 78	3 / 117
Nested-radical support	9	1 / 89	1 / 89	2 / 178	2 / 178

The nested-radical support takes one step at $d = 6, 12$ and two at $d = 30, 60$. Its summed trace allowances are $89, 89, 178, 178$. These bounds exceed the decentered support's $39, 78, 78, 117$ here: higher order alone does not establish a drawing-work advantage. The inventories are not minimum costs or runtime measurements. The iteration counts are high-precision numerical checks, not interval or floating-point certificates. The reproducible formulas and threshold errors are supplied in the comparison supplement. Section 6 separately counts the power-free direct enclosure.

6 Direct root enclosures for real degrees

The integer power stage of the iteration protocols can be avoided by constructing the root directly. The input is a positive real length X and a given real degree $p \geq 1$; the output is an approximation to $r = X^{1/p}$ with a prescribed relative error. A real input length is assumed to be available: this does not assert that every real number is constructible from a unit segment. The cases $X = 1$ and $p = 1$ are immediate. If $0 < X < 1$, apply the construction to $Y = 1/X$ and invert the resulting enclosure. We first treat $X > 1$.

6.1 One geometric mean per subdivision

Set $\theta = 1/p$, $a = 1$, $b = X$ and $\lambda = \theta$. In this section a, b are moving scalar rail values and r denotes the direct root, rather than the reciprocal-state target of the earlier sections. The invariant is

$$0 < a < b, \qquad 0 \leq \lambda \leq 1, \qquad r = a^{1-\lambda} b^{\lambda}. \tag{22}$$

If $\lambda = 0$ or 1, return the corresponding endpoint exactly. Otherwise construct $m = \sqrt{ab}$ with the classical mean-proportional semicircle [4] and use

$$(a, b, \lambda) \longmapsto \begin{cases} (a, m, 2\lambda), & \lambda \leq \frac{1}{2}, \\ (m, b, 2\lambda - 1), & \lambda > \frac{1}{2}. \end{cases} \tag{23}$$

Both branches preserve (22). The endpoint test is repeated after updating the weight. Thus p determines the choices and the weight, but never the number of factors in a power chain. After n subdivisions,

$$\frac{b}{a} = X^{2^{-n}}, \qquad z := \frac{b}{a} - 1 \geq 0. \tag{24}$$

The algorithm uses no logarithm, exponential or evaluation of s^p.

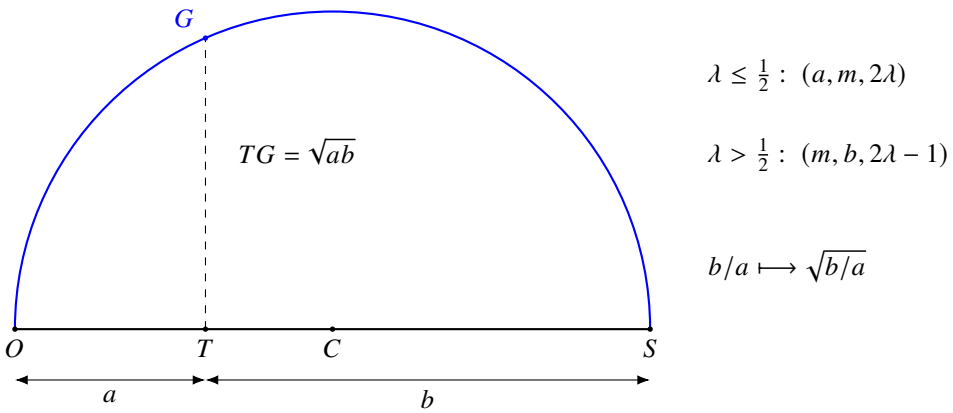


Figure 4: A subdivision constructs one mean proportional. The semicircle has diameter $OS = a + b$, center C, and junction $OT = a$. The same device is rotated onto the vertical rail in the counted protocol. The figure uses $a = 1.2$, $b = 2.8$ only to show the incidence.

6.2 A fifth-order final enclosure

The harmonic–arithmetic readout of the companion gallery [5] is

$$L_2 = \frac{ab}{(1-\lambda)b + \lambda a}, \qquad U_2 = (1-\lambda)a + \lambda b.$$

Weighted arithmetic–geometric mean bounds give $L_2 \le r \le U_2$, with

$$\frac{U_2}{L_2} - 1 = \lambda(1-\lambda)\frac{z^2}{1+z} \le \sinh^2\left(\frac{\log X}{2^{n+1}}\right).$$

Its relative width is quadratic in z. A single rational readout of degree two in numerator and denominator gives a fifth-order width. The readout is the diagonal $[2/2]$ Padé approximant of $(1+z)^t$ at zero [6, 7]. For $0 \le t \le 1$ define

$$P_t(z) = 12 + 6(2+t)z + (t+1)(t+2)z^2, \tag{25}$$

$$Q_t(z) = P_{-t}(z), \qquad \mathcal{P}_t(z) = \frac{P_t(z)}{Q_t(z)}. \tag{26}$$

All required denominators are positive for $z \ge 0$. Put

$$L_5 = a\mathcal{P}_\lambda(z), \qquad U_5 = \frac{b}{\mathcal{P}_{1-\lambda}(z)}. \tag{27}$$

Proposition 6.1 (Real-degree enclosure and finite termination)**.** *For $X \ge 1$, any given real $p \ge 1$, and every finite subdivision depth,*

$$L_5 \le X^{1/p} \le U_5, \qquad \frac{U_5}{L_5} - 1 = \frac{\lambda(1-\lambda)(2-\lambda)(1+\lambda)z^5}{P_\lambda(z)P_{1-\lambda}(z)} \le \frac{z^5}{256}. \tag{28}$$

Choose a positive rational δ with $\delta^5/256 \le \varepsilon$. Stop when $b \le (1+\delta)a$, and return L_5. The procedure terminates and its relative error is at most ε. Here $\varepsilon > 0$. The sufficient number of subdivisions depends on X and ε, and can be chosen independently of p.

Proof. For $0 \le t \le 1$, direct differentiation gives

$$(1+z)(P_t'Q_t - P_tQ_t') - tP_tQ_t = -t(1-t^2)(4-t^2)z^4 \le 0.$$

Consequently $\log \mathcal{P}_t(z) - t\log(1+z)$ is nonincreasing from zero, so $\mathcal{P}_t(z) \le (1+z)^t$. Apply this also at $1-t$ and use $(1+z)^t(1+z)^{1-t} = 1+z$ to obtain the upper bound. Multiplication gives the equality in (28). Writing $u = t(1-t) \le 1/4$, its coefficient is $t(1-t)(2-t)(1+t) = u(2+u) \le 9/16$. Since $P_t, P_{1-t} \ge 12$, the width is at most $z^5/256$.

At the stopping test $0 \le z \le \delta$, so $r/L_5 - 1 \le \varepsilon$ and $(r - L_5)/r \le \varepsilon/(1+\varepsilon)$. Equation (24) implies $z \to 0$, hence finite termination. For analysis only, any nonnegative integer n satisfying

$$2^n \log(1+\delta) \ge \log X$$

suffices. Equivalently, $X \le (1+\delta)^{2^n}$ can be checked with exact rational arithmetic when the inputs are rational; no logarithm is part of the geometric construction. If $0 < X < 1$, an enclosure for $1/X$ becomes $[1/U_5, 1/L_5]$ after inversion, with the same relative width. □

It is sufficient to draw only L_5; the analytic upper bound certifies its accuracy without constructing a second Padé quotient. For $0 < X < 1$, constructing and inverting this lower bound for $1/X$ gives an upper approximation of the original root with relative error at most ε. The width has order five in the rail gap z. This is not a claim that the subdivision is a fifth-order fixed-point iteration: each subdivision halves the logarithmic rail gap.

6.3 A complete, conservative drawing inventory

Use the extended vertical rail $R(q) = (W, H(1-q))$, its origin $R(0)$, and the prepared top centers of Section 2. The input lengths X, p are given on the rail. Use the convention of Section 2: intersections of drawn objects, fixed preparation and decisions are uncharged, while $P_{\parallel} = 2J + 3C_{\mathrm{arc}}$. Preparing the constants, choosing a branch and setting a compass radius are treated as in that convention. This is a mobile-work upper bound, not a bound on physical drawing size or an optimum.

The following macros make the count explicit.

- A sum or nonnegative difference of two rail values uses one compass arc on the existing rail, with the prescribed orientation. A zero result reuses $R(0)$.
- Multiplication uses the finite five-join macro of Section 2.2. Division also costs at most five joins. For $b \neq 1$, construct $V = CR(b) \cap OA = \mathcal{U}_{1/b}$, project $R(a)$ through V to $L(a/b)$, then return through $\mathcal{U}_{1/2}, \mathcal{U}_{1/4}, \mathcal{U}_{1/2}$. The successive values are $2a/b, a/(2b), a/b$. For $b = 1$, reuse $R(a)$.
- Multiplication by a prepared positive constant uses two joins, as in Section 2.2.
- A mean proportional uses $2J + P_{\parallel} + 3C_{\mathrm{arc}}$: form $R(a + b)$ with one arc; form $R((a + b)/2)$ with the two-join fixed scaling; draw the semicircle through $R(0)$; draw the horizontal through $R(a)$ with one parallel macro; transfer the resulting length to the rail with one arc. These are three arcs in total.

The updated weight uses the fixed scaling 2λ and, on a right branch, one subtraction of 1. Thus a subdivision, including the next weight, costs at most $4J + P_{\parallel} + 4C_{\mathrm{arc}}$.

For the final lower readout, use the identity

$$L_5 = a\left(1 + \lambda\frac{6z(2+z)}{Q_\lambda(z)}\right), \quad Q_\lambda = 12 + 6(2-\lambda)z + (1-\lambda)(2-\lambda)z^2. \tag{29}$$

An evaluation schedule uses seven general products, two divisions (including b/a), two fixed scalings by 6, and seven sums or differences. The seven products form $(1-\lambda)(2-\lambda)$, z^2, $(2-\lambda)z$, $(1-\lambda)(2-\lambda)z^2$, $z(2+z)$, the weighted quotient, and the final scaling by a. Its cost is at most $49J + 7C_{\mathrm{arc}}$.

Operation	J	$P_{\parallel}$	C_{arc}
Initial weight $\lambda = 1/p$	5	0	0
n subdivisions and weight updates	$4n$	n	$4n$
At most $n + 1$ stopping tests	$2n + 2$	0	0
One final lower Padé readout	49	0	7
Total, parallel macros expanded	$8n + 56$	0	$7n + 7$

In particular, at most $15n + 63$ elementary traces suffice for $X > 1$. Endpoint exits can only reduce this bound. For $0 < X < 1$, add the two five-join inversions for the input and output, giving $15n + 73$ traces. Constructing both endpoints would require additional work and is not included in this single-output count. These new macro counts are a written incidence inventory, not a Lean cost theorem.

6.4 Depth and precision

Table 5 gives sufficient worst-case depths for $X = 2$, uniformly for all real $p \geq 1$. The gallery column uses its harmonic–arithmetic readout; the new column uses (27). The table compares enclosure widths, not electrical or floating-point errors.

Table 5: Sufficient subdivision depths at $X = 2$, uniformly in the given real degree $p \geq 1$. The last column includes initialization, stopping tests and one final readout.

Relative width target	Gallery depth	Padé depth	New trace bound
10^{-6}	9	2	93
10^{-8}	12	4	123
10^{-12}	19	6	153
10^{-15}	24	8	183

The Padé depths use, respectively, $\delta = 19/100, 3/40, 3/250, 3/1000$. The rational inequalities $\delta^5/256 \leq \varepsilon$ and $(1+\delta)^{2^n} \geq 2$ independently verify their sufficiency. For $p = 37.5$, four subdivisions yield $a = 1$, $b = 2^{1/16}$ and $\lambda = 32/75$. The resulting root lies, for example, between 1.0186558096 and 1.0186558103; these rounded endpoints can be checked exactly by raising them to the integer power 75 and comparing with 4.

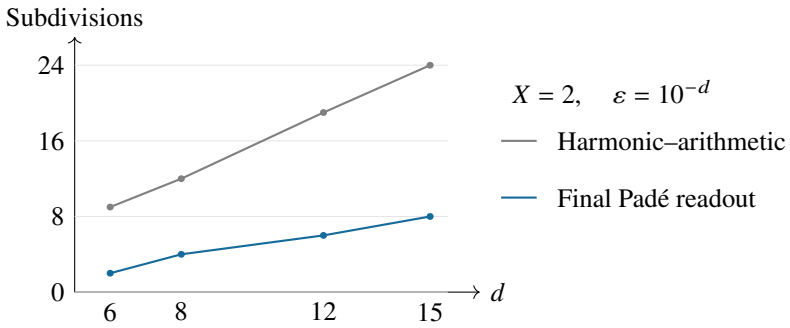


Figure 5: Sufficient worst-case depths for Table 5, uniformly in $p \geq 1$. Points are verified depth certificates; connecting segments guide the eye and do not represent fractional subdivisions. The final readout cost is included in the table's trace bound.

At $X = 2$ and $\varepsilon = 10^{-8}$ the construction has the bound $88J + 35C_{\text{arc}} = 123$ traces. This is a complete conservative inventory under the stated preparation convention, not a comparison of runtime or physical arc and join costs.

The direct method accommodates a given real degree without evaluating a residual power. It is a separate root calculator. Its finite-depth output is a certified approximation in exact geometry; arbitrary positive roots are not asserted to be exactly constructible in finitely many operations.

7 Attribution and algebraic approximation

High local order and geometric iteration are not new general principles. Traub organizes classical iteration formulas and their error terms [8, 9]. Gander discusses Halley's method and radical corrections [10]; Scavo–Thoo analyze its geometry [11]. Melman connects geometry with global convergence, and Amat–Busquier–Gutiérrez develop geometric constructions for nonlinear iteration [12, 13]. Here AK is Newton under inversion and centered AD is Halley, realized in the shared rectangle; their scalar formulas are classical.

The decentered and nested supports are algebraic corrections defined by explicit positive radicals. Algebraic approximants generalize rational approximation and belong to an established broad methodology [3, 14]. The two-radical coefficient matching is positioned within that methodology. Section 4 proves a degree-four distinction at the cubic specialization, without claiming historical priority. Appendix C gives a specified quadratic comparison for the decentered support and explains the scope of coordinate-change arguments.

For clarity, put $S(t) = (p-2)(t^2+1) + (10p+4)t$. Eliminating the decentered positive radical gives

$$(p-1)^2 S(t)(v-1)^2 - 12p(tv-1)^2 = 0. \tag{30}$$

The explicit radical in (4) selects the branch; the polynomial alone loses sign information and has a double root at $t = 1$.

The subdivision mean proportional is classical Euclidean geometry [4]. Its final rational readout is an ordinary Padé approximation [6, 7]; the enclosure and explicit width are proved in Section 6. This final readout is not a fixed-point iteration order. Counting particular constructions follows the concerns of geométrographie and nomography [15, 16].

The contribution evaluated here is the stated rectangle realizations, domains, scalar convergence proofs and protocol bounds. We do not claim priority over all equivalent formulas, a minimum trace count, or electrical speedup. The exact geometric construction, the scalar real iteration and a physically inaccurate drawing are distinct objects.

8 Evidence and reproducibility

The nested-radical results have two project Docker/Lean audits totaling 104 theorems: 38 cubic, 14 real-parameter certificates, 44 real-parameter analysis and 8 degree-extension theorems. They use only `propext`, `Classical.choice` and `Quot.sound`, with no admitted proof. The sources, pinned toolchain, hashes and audit records accompany the paper [17–19].

Lean module	Certified scope
`NestedRadicalCubic`	Cubic global convergence and exact relative-error order nine, coefficient $7/648$, for every positive target.
`NestedRadicalRealCertificate`	Exact polynomial identities and positive-coefficient certificates.
`NestedRadicalReal`	Original formula for every real $p > 2$; support equivalence and small-degree counterexample.
`NestedRadicalAllDegrees`	Degree doubling, visible starting-value preservation and convergence for every $p \neq 0$.
`GeometricSubdivision`	Retained scalar interval invariant, real-degree rational bracket and width; stored seven-declaration audit.

The general local order-at-least-nine expansion is a written analytic proof with exact coefficient checks. The new method's illustrative example is checked independently at 110 decimal digits without using the unknown cube root to place points. Its complete inventory has 64 exact rational macro checks and 60 independent full line/circle evaluations at 160 digits, including support placement and report; four additional root-stop cases are checked. Neither is represented as a Lean geometry or floating-point theorem. The decentered global argument and the classical rectangle derivations are written proofs; their geometric costs, including Proposition 5.1, are written inventories rather than a Lean cost semantics.

The nested verifier checks coefficient identities, the cubic positive certificate, the degree-four specialization, eight exact local series and 80 residual probes at 140 digits. The six-row comparison is independently rerun at 250 and 500 digits, with the same initial state and relative-error stopping rule. Finite numerical probes supplement proofs and do not certify rounded implementations.

The subdivision verifier checks three exact polynomial identities, 188 rational rail incidences, four rational depth certificates, the outward-rounded $p = 37.5$ example and 256 numerical enclosures at 100 digits [20]. The retained subdivision Lean audit is not described as newly compiled. The complete source package contains `build.sh`, `verify.py`, the independent comparison script and validation records. The gallery contains companion interactive views; it does not certify finite drawing error.

9 Conclusion and open questions

Changing the support in the same Pandrosion rectangle yields a progression from classical orders one to four to a globally monotone fifth-order radical and a nested-radical support with exact cubic order nine. The latter's original scalar family converges globally for real $p > 2$, with an explicit degree-doubling extension to every $p \neq 0$. Lean verifies those global results and the exact cubic local limit; the general local calculation gives order at least nine by a separate analytic proof.

The matched cubic comparison uses the same initial state for all six constructions. The nested support now has a complete uniform bound of $10p + 59$ traces per step, or 89 at $p = 3$. Its $89, 89, 178, 178$ summed bounds do not improve on the specified decentered protocol in this example, although its iteration counts are lower. The power-free subdivision method addresses a separate problem, giving a finite-depth real-degree enclosure with relative width at most $z^5/256$ and the bound $15n + 63$ for $X > 1$. Residual powers remain necessary in the fixed-point iterations.

The remaining questions are to determine and formalize the general ninth-order coefficient, shorten the nested-support inventory and formalize its cost semantics, analyze finite drawing and floating-point conditioning, and compare exact formulas with earlier algebraic approximants under specified transformations. Historical priority, optimality and hardware performance are not established by the scalar convergence theorems.

A Classical rectangle supports

A.1 AK and centered AD: orders two and three

AK: a fixed line. Prepare $K = (W(1 - X/p), 0)$. The support AK has $\kappa = p$, so $\mathcal{T}_{\mathrm{AK}}(s) = ps/(p - 1 + t)$, Newton under inversion. It has order two and converges from every positive start.

AD: a centered length transfer. Prepare

$$O_{\mathrm{AD}} = \left(W - \frac{2W}{p-1},\ H - \frac{H(p+1)}{X(p-1)} \right).$$

Transfer $AE = Hs^p$ downward from this center to D_{AD} and join AD_{AD}. Then $\kappa = [p + 1 + (p - 1)t]/2$, giving

$$\mathcal{T}_{\mathrm{AD}}(s) = s\frac{p + 1 + (p - 1)t}{p - 1 + (p + 1)t}. \tag{31}$$

This Halley map has cubic order and global monotone convergence. Earlier geometric interpretations appear in [10, 11]; here the rail encoding and readout are shared with AK.

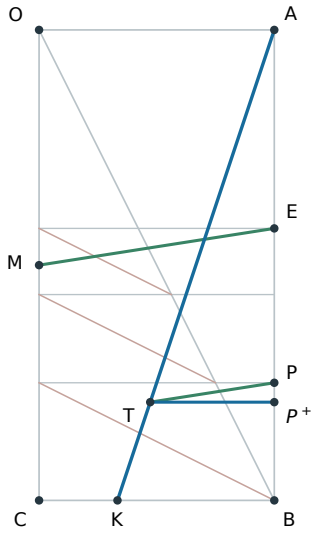


Figure 6: Pandrosion AK, order two. The fixed support AK replaces MA while the triangle chain and green correction direction are retained. Here $p = 3, X = 2, s = 3/4$.

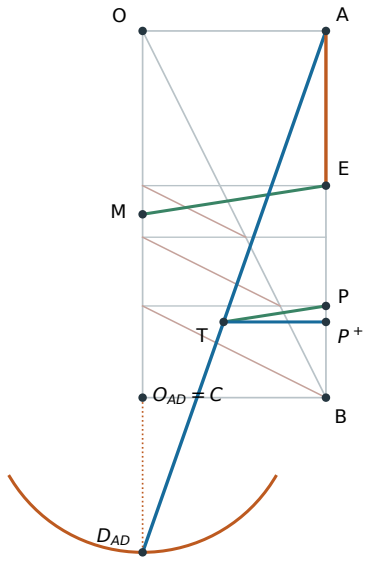


Figure 7: Pandrosion AD, order three. Transfer the orange length AE onto the prepared vertical through O_{AD}; the lower arc point determines AD_{AD}. Here $O_{\mathrm{AD}} = C$.

Both drawings retain the triangle stage, green transport and horizontal readout; only the support changes.

A.2 Projective AD: order four

Two joins supply the support. Prepare

$$\ell:\ y = H\left(1 + \frac{5p-1}{(p+1)X}\right), \quad Z_{\mathrm{proj}} = \left(W\left(1 + \frac{5p-1}{2p(2p-1)}\right), H\left(1 + \frac{p+1}{(2p-1)X}\right)\right).$$

Set $D_{\mathrm{proj}} = Z_{\mathrm{proj}}E \cap \ell$ and join AD_{proj}. Substitution in (3) gives

$$\mathcal{T}_{\mathrm{proj}}(s) = s\frac{2p((2p-1)t + p + 1)}{(p+1)t^2 + 2(2p-1)(p+1)t + (2p-1)(p-1)}. \tag{32}$$

The direct-root correction is Padé $[2/1]$; the reciprocal-state factor displayed here is $[1/2]$. Denote its positive denominator by $Q_{\mathrm{proj}}(t)$. Direct differentiation gives

$$\mathcal{T}'_{\mathrm{proj}}(s) = -\frac{2p(p^2-1)(2p-1)(t-1)^3}{Q_{\mathrm{proj}}(t)^2}.$$

The map has its global maximum a at $s = a$. Also $\mathcal{T}_{\mathrm{proj}}(s)/s - 1 = (1-t)((p+1)t + 5p - 1)/Q_{\mathrm{proj}}(t)$. Thus a first step lands at or below a, and subsequent steps increase to a. Integrating the derivative at a gives exact order four. The archival companion gives further dynamics and the original triangle-chain realization [21].

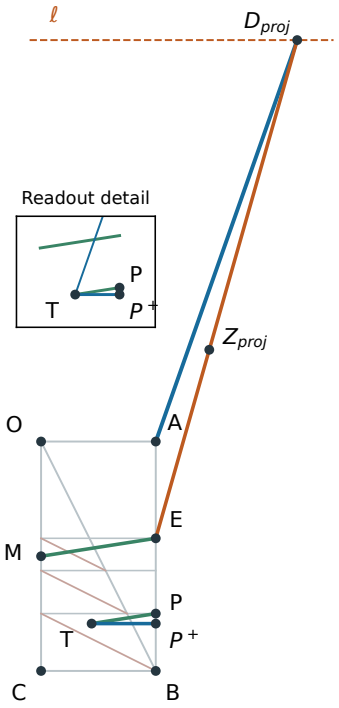


Figure 8: Projective AD $[2/1]$, order four, at the same cubic state. The orange line $Z_{\mathrm{proj}}E$ meets the prepared horizontal ℓ at D_{proj}. The blue support AD_{proj} receives the green correction. The inset separates P, T and P^+.

B The linear MA reference

MA provides a baseline, not an additional accelerated method. Write its lower-right endpoint as $A_{\mathrm{lin}} = B$. The parallel to ME through P meets MB; a horizontal returns the next state. For $s \neq 1$,

$$\mathcal{T}_{\mathrm{MA}}(s) = 1 - \frac{X-1}{XS_p(s)}, \qquad S_p(s) = \sum_{j=0}^{p-1} s^j.$$

For $X > 1$ this scalar map is increasing and $\mathcal{T}_{\mathrm{MA}}(s) - s = (1 - Xs^p)/(XS_p(s))$. It therefore converges monotonically to a from every positive start, with the removable scalar extension at $s = 1$. Its derivative at a is $1 - pa^{p-1}/S_p(a) \in (0, 1)$: convergence is linear.

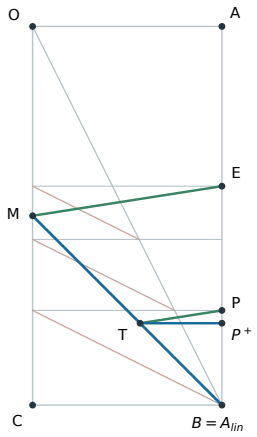


Figure 9: Pandrosion MA, the linear reference construction, at $p = 3$, $X = 2$, $s = 3/4$. Thin lines construct the power by similar triangles. The green parallel meets the blue support MA_{lin}; the horizontal gives P^+. The historical endpoint A_{lin} is the lower-right vertex B in this paper. At $s = 1$ the geometric intersection is coincident despite the removable scalar extension.

The notation $S_p(s)$ denotes only this geometric sum.

C Equivalence of the decentered correction

The question whether the arc is a known formula "up to conjugacy" needs a specified class of coordinate changes. The comparison depends on the allowed transformations. Let $P_p(t, v)$ denote the left side of (30).

Homographic changes. For each real $p > 2$, the correction $\Phi_{\mathrm{arc}}(t)$ is not rational in t. Indeed,

$$\operatorname{disc}_t S = 96p(p+1), \qquad S(0) = p - 2 > 0,$$

so S has two simple, nonzero complex roots and is not a square in $\mathbb{C}(t)$. If the correction were rational, then

$$\sqrt{S(t)/(12p)} = \frac{1 - t\Phi_{\mathrm{arc}}(t)}{(p-1)(\Phi_{\mathrm{arc}}(t) - 1)}$$

would be rational, a contradiction. Invertible homographies of the input or output preserve rationality. For integer $p \geq 3$, $S(x^p)$ also has only simple roots; hence the full state map $x\Phi_{\mathrm{arc}}(x^p)$ is nonrational and cannot be Möbius-conjugate to a rational state map, even as a germ at $x = 1$. This state-map exclusion is asserted only for integer degrees.

A specified quadratic normalization. A diagonal type $(2, 2, 2)$ quadratic Hermite–Padé (Shafer) form for $f(t) = t^{-1/p}$ requires polynomials of degree at most two with

$$A(t)f(t)^2 + B(t)f(t) + C(t) = O((t-1)^8);$$

this is the standard degree/contact convention [22, eqs. (1.4)–(1.6)]. Direct substitution into our quadratic instead gives, with $u = t - 1$,

$$P_p(1+u, (1+u)^{-1/p}) = \frac{(p-2)(p-1)^2(p+1)(3p+1)}{60p^4}u^6 + O(u^7).$$

Thus it has contact six, not eight. The coefficients of P_p have no common polynomial factor, and its discriminant in v is $48p(p-1)^2(t-1)^2S(t)$, so it is irreducible over $\mathbb{C}(t)$. Any polynomial of the same degree bounds defining this branch is therefore a constant multiple of P_p. This excludes that particular diagonal approximant in these coordinates, without excluding other algebraic families or transformed targets.

Local analytic conjugacy. Distinct formulas may still share a local analytic normal form. Böttcher's theorem conjugates a superattracting analytic germ $F(e) = ce^m + O(e^{m+1})$, $c \neq 0$, to its monomial normal form [23]. This does not supply a finite incidence protocol or transport a global domain and drawing inventory. The explicit construction and proved dynamics are therefore distinct from a claim of a new local analytic conjugacy class.

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